Nicholas Pickering

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Professional Summary

Software engineer with extensive experience designing, optimizing, and maintaining large-scale distributed systems using C and C++. Strong academic and industry background in applying complex data structures and algorithms to drive system performance. Proven track record of taking end-to-end ownership of mission-critical infrastructure, with a deep focus on system health, advanced debugging, diagnosis and resolution, and rigorous software test engineering.

Experience

Simulator Engineer — **Westinghouse Electric Company LLC.**
Full-time — June 2024 – Present

- Architected and shipped high-performance distributed simulations for large-scale nuclear power plant infrastructure ($30M+ projects), utilizing C, C++, and FORTRAN to model complex system behaviors.
- Led diagnosis and resolution efforts for mission-critical operations, aggressively debugging complex distributed networks and maintaining rigorous code and system health standards.
- Implemented comprehensive software test engineering practices to de-risk live software deployments, validating control logic and performance offline before executing site installations in global environments.
- Engineered automated network configurations that eliminated systemic bottlenecks, improving overall system health and saving $15,000 in configuration costs per project.

Software Development Intern — **MOSIMTEC**
Full-time — Mar 2022 – Dec 2022

- Developed predictive models and executed large-scale data analysis using Python, leveraging visualization tools to communicate actionable insights and system behaviors to enterprise clients.
- Implemented statistical algorithms to evaluate and improve the performance of high-throughput operational flows, driving configuration optimization and operational efficiency.

Projects

CNN Training for Image Classification Executed in HPC Environment — **2025**

- Analyzed CNN architectures for image classification on datasets of increasing complexity, implemented in Python with PyTorch.
- Designed for execution on HPC clusters using SLURM for job scheduling and increasing parallelism to determine problem Strong and Weak scalability.

Education

BS in Modeling and Simulation Engineering — **2019 – 2024**
Old Dominion University — GPA: 3.8/4.0
Concentration in Advanced Simulation Techniques (Artificial Intelligence, Distributed Computing, Visualization)

Technical Skills

- **Programming Languages:** C, C++, Python, Java, Rust, SQL
- **Systems & Architecture:** Large-Scale Distributed Systems, System Health & Monitoring, Diagnosis and Resolution, TCP/IP & UDP Protocols, High-Performance Computing (HPC)
- **Data & Algorithms:** Data Structures, Algorithm Design, Large-Scale Data Analysis (Pandas, NumPy, Polars), Visualization Tools (matplotlib), Statistical Modeling
- **Testing & Code Quality:** Software Test Engineering, Advanced Debugging, Version Control (Git)
- **AI & Frameworks:** PyTorch, scikit-learn, Model Deployment, Predictive Modeling, Machine Learning Integration

Awards

National Training and Simulation Association (NSTA):

- Graduate Scholarship Award in Modeling & Simulation (2024)